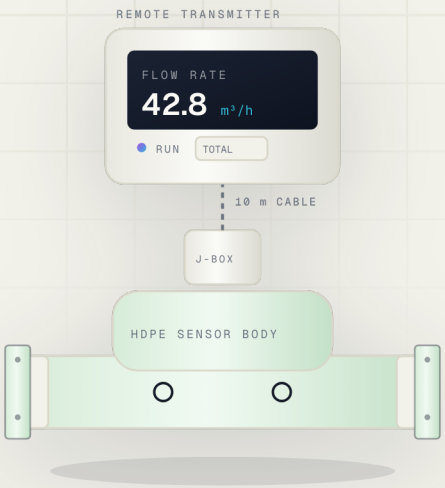
VEPOLINK
MONITORING PURITY

PROCESS INSTRUMENTATION × FLOW MEASUREMENT

RUST-FREE FLOW MEASUREMENT FOR CORROSIVE LIQUIDS

EcoFlow Flowmeter *Plastic-Body Electromagnetic*

A molded HDPE body electromagnetic flowmeter for small and mid bores — lightweight, entirely rust-free, and built for corrosive and aggressive conductive liquids at a fraction of a metal body's cost.



10–200 mm
NOMINAL BORE RANGE

HDPE
ENTIRE BODY CONSTRUCTION

±1 %
ACCURACY OF MEASURED VALUE

FARADAY ELECTROMAGNETIC INDUCTION | **REMOTE** TRANSMITTER ASSEMBLY |
VEPOLINK FIELD-PROVEN BUILD

10–200mm
NOMINAL BORE RANGE

±1%
ACCURACY OF MEASURED VALUE

0 corrosion
ENTIRE PLASTIC BODY CONSTRUCTION

5 kg/cm²
RATED WORKING PRESSURE

01 OVERVIEW

Electromagnetic flow, *without the metal.*

An entirely plastic-body electromagnetic flowmeter for pipe sizes 10 mm to 200 mm, giving performance in line with metal-body meters at lower weight and cost. The molded HDPE body is rust-free and suitable for virtually all conductive liquids, including corrosive and aggressive chemistries that would attack a metal sensor over time.

A remote transmitter assembly keeps the electronics off the wetted sensor, with hermetically sealed, welded coils and field-interchangeable electronics for easy upkeep in the field.

REPORTED CHANNELS

- INSTANTANEOUS FLOW RATE
- FORWARD TOTALISER
- REVERSE TOTALISER
- NET TOTALISER
- 4-20 MA OUTPUT
- FLOW DIRECTION
- EMPTY PIPE DETECTION
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Entirely Plastic Body

A molded HDPE sensor body is rust-free by construction, so there is no metal wetted surface to corrode on aggressive liquids.

No Pressure Loss

A full, unobstructed bore keeps the flow path open, adding no restriction and no pressure drop to the line.

Light Weight

A plastic body is markedly lighter than a metal sensor of the same bore, easing handling and support on small-bore runs.

Remote Transmitter

A remote transmitter assembly with field-interchangeable electronics keeps the display and controls clear of the sensing point.

02 MEASUREMENT PRINCIPLE

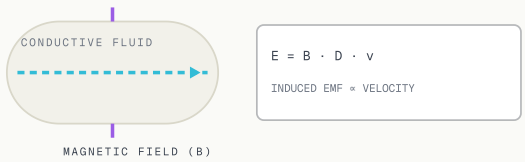
The same physics, *a lighter body.*

The measurement is unchanged from a metal-body electromagnetic flowmeter — only the material carrying the coils and electrodes is different, so the body itself is what resists corrosion rather than a separate liner.

FARADAY

Electromagnetic Induction

MAGNETIC COILS · CONDUCTIVE FLUID · INDUCED EMF



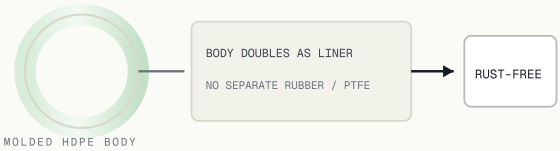
A magnetic field (B) is generated across the tube bore. A conductive liquid moving through it at velocity (v) induces a voltage (E) directly proportional to flow velocity.

`E = B · D · v`

CONSTRUCTION

Molded Body as Liner

HDPE BODY · NO SEPARATE LINER



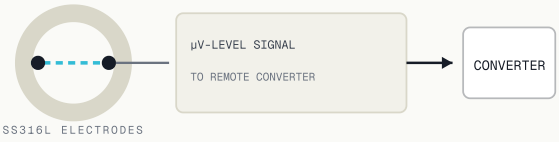
The HDPE body itself is the non-conductive flow-tube wall — there is no separate rubber or PTFE liner to wear, age or disbond.

`Corrosion resistance by material, not coating`

PICKUP

Electrode Signal Pickup

SS316L ELECTRODES · MV-LEVEL SIGNAL



A pair of SS316L electrodes, molded flush with the HDPE bore, pick up the induced voltage and carry it to the remote converter.

`Signal ∝ flow velocity`

ASSEMBLY

Remote Transmitter

10 M CABLE · FIELD-INTERCHANGEABLE ELECTRONICS



The sensor's signal travels over a standard 10 m cable to a remote transmitter, which converts, displays, totalises and reports it onward.

`Sensor and electronics kept apart`

03 SENSOR CONSTRUCTION

Built for *aggressive liquids.*

Every wetted part is chosen so the meter can sit in corrosive or aggressive process liquids indefinitely, without the slow metal fatigue a conventional sensor body would face.

STEP 01

Molded HDPE body

The flow tube and sensor housing are molded as one rust-free plastic body, flanged for inline mounting.

STEP 02

Field coils energised

Hermetically sealed, welded coils generate a controlled magnetic field across the bore, exactly as in a metal-body meter.

STEP 03

Electrodes sense EMF

SS316L electrodes molded into the HDPE wall pick up the µV-level induced voltage from the passing liquid.

STEP 04

Remote conversion

The signal travels over cable to the remote transmitter for display, totalising and output.

WHERE PLASTIC BODY WINS

On acids, alkalis and other aggressive chemistries, a metal sensor body depends on its liner holding up over years of service. An entirely plastic body removes that dependency, and removes the weight and cost of a metal housing on small and mid bores.

WHERE IT DOES NOT FIT

Rated to 5 kg/cm² and 55 °C, and bores to 200 mm — outside that envelope, or on higher-pressure, higher-temperature or very large-bore lines, the metal-body inline flowmeter is the better fit.

WHY THIS MEASUREMENT MATTERS

Accurate flow measurement on corrosive dosing, effluent and chemical process lines is often the hardest place to keep a sensor alive. A rust-free body answers that directly, at a lower installed cost than a lined metal alternative for small and mid-size lines.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

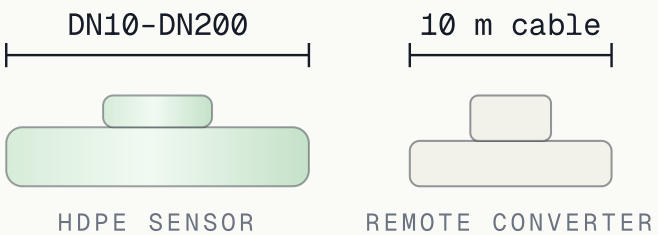
Parameter	Volumetric flow, conductive liquids
Working Principle	Faraday's law, electromagnetic induction
Nominal Diameter	10 - 200 mm
Measuring Range	0.2 - 12 m/s, bidirectional
Accuracy	±1% of measured value
Repeatability	±0.2% of span
Working Pressure	5 kg/cm²
Working Temperature	Up to 55 °C

SYSTEM · BUILD · POWER

Display	2-line LCD, local units (m³, litre, gallon, ft³)
Transmitter	Remote assembly, standard, field-interchangeable electronics
Measuring Tube	HDPE
Sensor Housing	HDPE
Sensor Lining	Not applicable — body doubles as liner
Electrode Material	SS316L, standard
Output	4-20 mA, standard
Power Supply	12-60 VDC or 80-300 VAC/DC · solar powered option
Protection Class	Sensor IP65 (IP67 / IP68 option) · transmitter IP67
End Connection	Flange · ANSI 150 standard
Remote Cable Length	10 m, standard

INSTRUMENT DIMENSIONS

INDICATIVE SENSOR & REMOTE CONVERTER · NOT TO SCALE



05 INSTALLATION & UPKEEP

Light to install, *low to maintain.*

> The sensor's light weight and flanged plastic body make it easier to handle and support than a metal meter of the same bore. Installation and ongoing maintenance are **minimal** — there are no moving parts, no liner to inspect, and no corrosion to track.

01 · ORIENTATION

Always full bore

Install with the tube always full — vertical upflow or a low point in a horizontal run — to avoid false readings from a part-full pipe.

02 · SUPPORT

Light but supported

Lower weight eases handling, but the line should still be supported either side of the flanges per standard piping practice.

03 · GROUNDING

Bonded to the process

Ground rings bond the liquid to pipe potential even on a plastic body, keeping the electrode signal clean.

04 · CABLE RUN

Standard 10 m

The remote transmitter sits up to the standard 10 m cable length away — convenient for pits, trenches or tight chemical skids.

05 · PRESSURE & TEMPERATURE

Within rated envelope

Confirm the line stays within 5 kg/cm² and 55 °C — the limits of the plastic body construction.

06 · DIAGNOSTICS

Reported with the data

Empty-pipe detection, flow direction and fault status travel alongside rate and totaliser outputs.

SIZING A METER

Nominal diameter is selected to keep expected flow within the 0.2–12 m/s velocity window at the chosen bore. Confirm the liquid's chemistry, pressure and temperature sit inside the plastic body's rated envelope before specifying; our team can confirm fit from your process data.

COMMISSIONING

Commissioning covers grounding verification, an empty-pipe and zero check with the line full and static, and confirmation of output scaling against the plant's control or billing system — after which the meter runs unattended, with no consumables and no moving parts to service.

06 APPLICATIONS

Where it *works.*

Any electrically conductive liquid, especially corrosive or aggressive ones, on bores up to 200 mm — where a rust-free body and a lower installed cost both matter.

Chemical & Petrochemical	Pharmaceutical	Sugar & Beverage
Water & Wastewater	Mining Industries	Fertiliser Industries
Food & Drug Industries	Paper & Pulp	Dredging Industries

LOWER COST ON SMALL BORES

On the 10–200 mm range typical of dosing lines, utility branches and skid piping, a plastic body meter gives comparable accuracy to a metal-body meter at a lower material and installed cost — without trading away corrosion resistance.

A COMPANION TO THE FULL-BORE LINE

Larger bores, higher pressures or higher temperatures are better served by a metal-body inline flowmeter; the two share the same measurement principle and typically sit on the same plant alongside each other.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Faraday's law of electromagnetic induction — the same principle as a metal-body electromagnetic flowmeter. Only the body material differs.

Q Why choose a plastic body over metal?

An entirely HDPE body is rust-free and well suited to corrosive and aggressive conductive liquids, at lower weight and lower cost than a lined metal sensor — on bores up to 200 mm.

Q What sizes are available?

Pipe sizes from 10 mm to 200 mm, sized to keep expected flow within the 0.2–12 m/s velocity window.

Q What pressure and temperature is it rated for?

5 kg/cm² working pressure and up to 55 °C. Outside that range, the metal-body inline flowmeter is the better fit.

Q Does it cause any pressure loss?

No — the full, unobstructed bore adds no restriction and no pressure drop to the line.

Q Is the transmitter integral or remote?

Remote assembly as standard, connected by a 10 m cable, keeping the electronics clear of the sensing point.

Q Can it measure flow in both directions?

Yes, it is bidirectional, reporting forward, reverse and net totals from the same sensor.

Q What outputs are available?

A local 2-line LCD display and a standard 4–20 mA analog output for integration with plant control systems.

Q What liquids is it suitable for?

Virtually all electrically conductive liquids, including a wide range of corrosive and aggressive chemistries that the HDPE body resists without a separate liner.

Q What maintenance does it need?

Very little — there is no liner, no moving parts and no corrosion to track. Periodic grounding and zero checks per the site's audit schedule are typically all that's required.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

FLOW, MEASURED WITHOUT CORROSION

Know your flow. *Rust-free.*

Tell us the liquid, the bore and the line conditions, and we'll confirm fit and configure the sensor and transmitter around your process. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that sizes, specifies and commissions electromagnetic flowmeters. No call centre.
+91 99538 86178 →

02 **Request a sizing & configuration**
RECOMMENDED
Send your bore, liquid and line conditions — we'll confirm fit and spec the transmitter and output.
sales@vepolink.com →

03 **Book a demonstration**
See a running meter and the converter output with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM